ENGLISH

Chapter 7

Personal Growth

Thus far I've focused on protocols, and the science they're based on, for bolstering key pillars of your health and performance: sleep, exercise, nutrition, stress management, light exposure, as well as focus, motivation, and learning. You'll notice I did so mostly through the lens of biology, not psychology. Even when talking about mindsets or ways to generate a sense of urgency, the emphasis unabashedly has been on the biological underpinnings of what makes these protocols work. That reflects my training and interests, but as I mentioned in the introduction it's also an essential part of how these protocols work. They are customizable to your needs.

When you know how a machine works—how the parts fit together—you come away knowing which aspects you can modify. We, however, are not machines. We think and feel, and hope and fear, and more. That needs attention too. The quality of our lives is strongly driven by our emotional lives, our relationships to others and to ourselves.

After all, what good is it feeling vital and rested, focused and resilient, fit and capable, if we don't also enjoy our time with others, and with ourselves? We just can't overlook these aspects of life. Scores of data now tell us the obvious: that social connection is vital to our health and longevity.J Like all the other Old World primates, we have an immense need to care for and be cared for by others.Z Our nervous system evolved to crave and require these types of interactions we

consciously or unconsciously register as safety. Not just safety from harm, but also the sort of safety that makes us feel capable of seeking out challenges: healthy risk-taking in our work, new goals in our hobbies, going on adventures; all things that give us a sense that life, our life, matters. Equally important but far less discussed is our relationship to ourselves. That's an admittedly squishy term. What I mean is how we talk to ourselves, and how we think about ourselves— the decisions we make, the paths we choose.

In fact, this chapter could have been called "Protocols for Thinking." My goal here is not to get you to second-guess yourself. Quite the opposite. It's about clearing the clutter from your mind so you can think better. It's about moving forward on the right path and not mindlessly slaving away on the wrong one. What are thoughts anyway?

Marcus Aurelius famously said, "Our life is what our thoughts make it"4 and nearly identical statements were made by René Descartes, the Buddha,6 Ralph Waldo Emerson,Z William Shakespeare,g and in the Bible.2 Most people would agree their thoughts are a crucial feature of their experience of life but few ever stop to ask, "What are thoughts?" Most neuroscience textbooks don't even attempt to answer that question. They include chapters on sensation, perception, emotions, behaviors, even memories and dreams, but not a single textbook that I am aware of provides a good explanation of what thoughts are. Fortunately, researchers are starting to figure it out.

distributed networks. Which sensory area is activated first and which ones follow depends on what "seeds" the thought. In the above example it was the word cat but think about the extremely soft fur of a chinchilla. That may set off a thought that activates the touch area of your brain, and you might experience a simulation of touch, what neuroscientists call somatosensation. From there, the patterns of brain activity expand.

Regardless of where it starts, focusing on something—anything—causes your brain to call up memories, as well as any potential motor actions you might take in relation to it. In some, it also conjures up emotions related to that thing. This "simulation" aspect ofhuman brain

function is vitally important. Not just as a way to navigate our lives, but for how we eventually come to make sense of our different life experiences. It is among our most valuable mental resources and a key feature of what sets us apart from the other species on earth.

As Groh explained, "Humans don't hear or see any better than a macaque monkey. What we do have is more brain areas devoted to each of the same senses, which makes our brain capable of building up very elaborate simulations of what we perceive." In other words, the more we focus on something the richer the simulation of that thing becomes. That valuable mental resource is adaptive. It's what allows us to make predictions of "If A then B, and if B then maybe C" and so on. It's what allows us to imagine solutions to certain problems, and to build tools and technologies to overcome them.

It's what gave us electricity, planes, the internet, smartphones, and so much more.

Our brains can only run a certain number of simulations at once. Groh described an experiment she does that illustrates this, and why. She stops one of her Class lectures midway through and asks her students to try to think of a word totally unrelated to what they have been discussing. Sounds easy, right? Wrong.

Eventually the students come up with a word, but it takes significantly more time and effort than if she asked them to think of a word related to what they have been discussing. It's as if thoughts are roads: The more time you spend thinking about something, the farther you travel down that road. Sounds obvious, but is it true? Yes. Neuroimaging and direct recordings of brain electrical activity confirm that as thought-triggered simulations get more elaborate, they outcompete other simulationsn Neuroscientists call these attractor states, which is when brain activity increases in a certain set of circuits and as it does, activity in other circuits diminishes.&?

Incidentally, this phenomenon may be what led to the idea that humans only use a small percentage of our brain at any one time—which unfortunately made it seem like we were dumber than we could be if only we used more of our brains. Now that you understand a bit about how thoughts work, it should be obvious Why that's a silly idea. Who would want too many thoughts happening at once?! Thought competition and attractor states are vital, in fact. Groh used a great example to explain why.

Think about when you're driving and you say to someone, "Hold that thought for a second, I need to change lanes." You are reflexively limiting the amount of information coming in so you can

switch simulations—from talking and listening, to driving and navigating. You are allocating resources to different brain circuits. So maybe you're now asking, "What in the world does that mean for improving our relationship to ourselves?" Turns out: everything. Improving our relationship to ourselves is about being intentional with what thoughts we sow. Gone unchecked, our thoughts grow, and grow, and grow.

This can be a good or a bad thing for our mental and physical health and how we show up in the world. It's important you know that your thoughts occur downstream of your perceptions. It's hard to overstate how strong this link really is. The way we see our lives, the way we narrate our lives and the events of the world at large to ourselves, shapes the thoughts we have. Ifyou tell yourself that every time you try to do something new, you fail—well, your thoughts Will reflect that perception.

This realization is the first step in having control over them. So yes, I'm saying you can control your thoughts. Below I explain how. I promise none of what follows is about the power of positive thinking. I Will not be guiding you through a series of manifestations.

The first protocol in this section is, of course, a sciencesupported tool. The other protocols that follow are about resolving a tension that exists for many people, regardless of age or stage of life, which is the push-pull between "needing to perform outward roles in life" and "being true to oneself." That is one of the most common challenges I hear from people and it can produce varying degrees of confusion, conflict, and in some cases, crippling

paralysis. I've often been asked by people forty and younger who are not already wholeheartedly committed to a career path, "What should I study, do, or be?" By contrast, people Older than forty usually ask me about making meaning of their life. With all the books and news articles and podcasts out there about finding a sense of purpose, I started to wonder: How can science help us answer these questions? It turns out there is a way. In my opinion, it's been best addressed by James Hollis, an eighty-six-year-old Jungian psychologist with more than five decades of clinical experience treating patients of all ages and backgrounds.

He has helped them build their best life by taking specific steps daily. He's not a life coach, mind you. He's a credentialed therapist who has also authored twenty-two books on this subject. He was a guest on the Huberman Lab podcast in 2024, and I personally have used the tools he described then every day since our conversation to great benefit.J-3 They resolve the tension between "needing to perform outward roles in life" and "being true to oneself." Hollis's "Daily Protocols for Creating Your Best Life," as I refer to them in this chapter, are something I wish I had heard long before that conversation. They help us make decisions aligned with our interests, and reduce a lot of second-guessing that can arise when we consider such actions.

No matter your age or level of contentment with your life, I am certain these protocols Will be of great benefit for you too. Finally, I include a protocol related to resolving inner conflicts that stem from unresolved pain. Formal traurna therapies are beyond the scope

PERSONAL GROWTH PROTOCOL Replace Mind Wandering with Present-Tethered Awareness

WHAT TO DO

• Throughout your day, periodically ask yourself if your thoughts are focused on the present. For example: "Are my thoughts focused on what I am doing and where I am, or are they related to the past or future?" Be honest in your response. • If your thoughts are elsewhere, re-tether them to the present. Ask questions anchored in your current surroundings, such as these: "How cold or warm is it in this room? What does it smell like? How many people in line in front of me are on their phone?" That seemingly trivial line of inquiry returns your perceptions to the present because they can only be answered right here, and now.

HOW IT WORKS

Most people would like to be happier. Among the many benefits, a better mood allows us to access more focus and creativity, enhances our rational assessment of information, and improves countless objective health metrics.L4 On the flip side, psychologist Jamil Zaki's lab at Stanford has

well-being. Just as important as material wealth, if not more so. The 2010s also brought a growing interest in the neural and psychological changes associated with mindfulness and also meditation, both of which are intended to help with being present. These practices are by no means new; they had existed in Eastern cultures for thousands of years, experienced fleeting popularity in the US in the 1970s, and then became trendy again. Now some fifteen to twenty years later, mindfulness is a familiar term, and meditation is a common practice.

We also have a lot of data from lab studies and clinical trials showing that our level of "presence" is not just the outcome but a cause of happiness, both in the short and long term. Okay, so we should "be present." Got it. But the whole "be present" thing is another pretty squishy term. Even for scientists wishing to study our level of presence, defining it is no easy task. To shift away from catchy phrases like "Be Here Now" (I love that phrase by the way) and run actual experiments, capture data, and suggest actionable tools on the basis of those data, scientists needed to create an operational definition of presence.

They also needed to study it in the real world, which is not easy. Such studies were done by a few pioneering scientists to whom we owe a debt of gratitude, such as Richard Davidson, PhD, from the University of Wisconsin—Madison, and Matthew Killingsworth, PhD, and Dan Gilbert, PhD, from Harvard. In 2010, Killingsworth and Gilbert published what is now considered a landmark study in Science.20 First, they created an app to collect real-time information on participants' thoughts

and feelings as they went about their day. The app sent question prompts to people at random moments during waking hours, such as, "How are you feeling right now?" and "What are you doing right now?" Every prompt also included the question "Are you thinking about something other than what you're currently doing?" The researchers amassed a database of nearly 250,000 responses from around 5,000 people ages eighteen to eighty-eight from eighty-three different countries. Then they dove in. Killingsworth and Gilbert analyzed the responses from 2,250 adults, most from the US, to look for any connections between the specific activity they were doing, their level of presence with that activity, and their mood. Their results showed that for nearly 50 percent of our waking lives, our minds are elsewhere, thinking about times and places and things that have nothing to do with the present.

But the researchers also showed the firm connection between presence and happiness. When study participants were focused on specific activities and on their immediate physical surroundings, they were happier. Even when those specific activities were boring or unpleasant, that focus was Still a buffer against unhappiness. "A wandering mind is an unhappy mind," the researchers wrote in a summary of their study.2J But perhaps the most interesting finding from the study was the causative link between mind wandering and unhappiness. When it came to feeling happy, mind wandering was a much bigger deterrent than whatever they were doing.

It didn't even matter what they were thinking about. Being present during a less enjoyable

activity seemed to make people happier than daydreaming about pleasant things unrelated to the present activity. When we are focused on the present moment, we are happier. This is the power of present-tethered awareness. While that study has some obvious limitations —for a start, it's based on subjective self-reports— it is notable for a few reasons. First, it has a realworld context.

It drew from large sample sizes. And it has readily implementable findings. It tells us that one of the most powerful determinants of our current and future levels of "happiness" is giving what we are doing as much of our undivided attention as we can. Notice I didn't say all our attention. Fully focusing 100 percent of our mental attention can happen, and that can be great—we discussed ways to do just that in the previous chapter.

But it appears that the mere act of periodically devoting a bit more of your attention to the present can be of meaningful benefit to your level of happiness. The data tell us that striving to anchor your thoughts to the present forms the scaffolding of your well-being. So if you'd like to be happy, and you want your children to be happy, find ways to practice present-tethered awareness alone or together. The questions included in the protocol above are a good way to start. Of course, feeling sad, disappointed, angry, et cetera, is sometimes warranted.

It's also healthy.Z2 Suppressing negative feelings has consistently been shown to lead to poorer mental and physical health outcomes.23 The problem is when we get stuck in those low-mood states too long.

As that 2010 study discussed above indicates, people's minds were already wandering about 50 percent of their waking hours before social media became such a prominent—arguably even dominant—feature of modern life that it is now. Tech companies just figured out how to get people to devote a significant percentage of that time to scrolling on their social media platforms. My View is that, in moderation, meaning about an hour or so per day, engaging on social media platforms can be great for learning, social interaction, and entertainment. The key is to not use social media to blunt your presenttethered awareness. We are starting to see the damage to mental and physical health that this can cause" Put simply, scrolling can become just another form of mood-lowering mind wandering.

The algorithms are strong. Their superpower is to get you to use them for as long as possible. Your superpower is your ability to consciously redirect your thoughts to specific activities and your immediate physical surroundings. That's a skill worth developing. If the benefits of doing so for your mood and well-being are not incentive enough, perhaps the longevity benefits 1'11 describe next Will be.

One of the causes of aging is the shortening of our telomeres, small molecular structures that form protective caps on the ends of our chromosomes in each of our cells. In cell types that duplicate, such as our immune cells, as telomeres get shorter, they stop dividing. At that point they become what are called zombie cells, and they secrete inflammatory molecules that damage nearby cells. Preclinical studies suggest that this zombie cell—induced damage may be a

factor in aging and age-related illnesses such as cancer, dementia, and heart disease.Zé Conversely, having cells with longer telomeres is a biomarker of youthful, healthy, and (for cell types that replicate) division-competent cells. With that knowledge in hand, UCSF psychologist Elissa Epel and colleagues conducted a study where they measured the average telomere length of immune cells from women who scored either high, moderate, or low on measures of mind wandering.2-$ She discovered that the women engaged in the greatest amount of task-irrelevant thought throughout the day had telomeres that were on average two hundred base pairs (a base pair is a functional unit of DNA) shorter than the telomeres of women with low arnounts of mind wandering. We all lose about 50 telomere base pairs for each year of life, so the difference between being someone who is present and someone whose mind wanders was equivalent to about four years of additional aging. This was an especially

remarkable finding because the benefits appeared independent of variables known to impact longevity, such as body mass index and stress level. The takeaway: Make the effort to be present in your daily activities, tasks, and relationships.

Doing so provides an actionable way to improve your mental and physical health and, potentially, your lifespan. This doesn't mean you have to be a Luddite. Texting a friend, sending a work email, or updating a family schedule on your Phone can definitely be about the "here and now." So is scrolling for education or entertainment. The thing to be wary of is texting and scrolling or

doing anything out of boredom or avoidance. Ask yourself: "Am I numbing out or getting drawn in by things that I can't actually have an impact on?" If you answer yes, that's a good indicator that you are engaged in deleterious mind wandering. Retired Navy SEAL and two-time world-recordholding athlete Andy Stumpf talks about this in his book Drownproof.29 And he shared a tool that I have found immensely useful, even lifechanging. Here it is: Take a sheet of paper and draw a line down the middle from top to bottom. On the left side, list all the things that concern you at that moment.

On the right side, list all the things you can actually control, and what you can do to directly impact your life that day, right then and there. Try it. You'll be amazed at what you learn about your mental preoccupations and reflexes, and what it opens up in terrns of your ability to better direct your mind and actions. Doing this exercise makes you keenly aware of what really matters to you. And it gives you a heightened sense of agency around those things.

It places you in a great mental stance to lean into the important stuffin your life. It also makes time spent relaxing on the couch or doing nothing feel restorative again. It's a simple tool, but a super valuable one. Yes, it's a good idea to do it every few weeks or once a month, but it's quick: five minutes at most. Do it.

Getting Out of Your Head I am a fan of certain forms of structured introspection. I'm referring to journaling, therapy with a well-trained expert who offers useful points of action, or other deliberate, timed selfreflection done within a specific framework.

There are very good data to support those practices as tools for mental health.30 What none of us want, however, is to be stuck in our own heads and feeling anxious. Turns out, we don't have to. We just need to understand a bit more neuroscience. And then apply that knowledge. Your brain has cells that are grouped into circuits, which in turn are grouped into networks.

Think of a network sort of like a setting for your brain: You have sleep networks, feeding networks, language networks, math networks, and on and on. And you have a network that turns on when you do nothing, aptly named the default mode network, or DMN. It was discovered to be active when people were getting their brains scanned, in between moments where they were asked to focus on certain tasksn That alone wasn't terribly surprising. What was surprising was that the DMN is hyperactive in ruminators, people with cognitive impairments like Alzheimer's disease, and in depressed patients.32 More than a decade of studies now show that many antidepressants work in part by dampening activity in the DMN.3.3 Guess what? So does dancing, exercising, socializing with friends, drawing... essentially anything that requires your attention.34 That's the key takeaway about the DMN.

You can't just "turn it off" or even reduce its activity unless you do something to activate a competing brain network, such as the salience network—a different collection of circuits that, you guessed it, become active when you focus your attention on something external to you. So

while your DMN has important healthy functions, like simulating what other people are thinking or feeling, which is useful for some

PERSONAL GROWTH PROTOCOL Apply James Hollis's Guide to Creating Your Best Life

WHAT TO DO

Practice these two things every single day. 1 Apply the Shut Up. Suit Up. Show Up motto to your everyday life. These are science-based tools to help bring your outward and inner lives into alignment. • Shut Up: Don't whine, instead incorporate a daily gratitude practice into your routine. • Suit Up: Use a task-switching protocol to move between tasks associated with your outward roles. • Show Up: Fulfilling our outward roles moves our life forward. That is what gives you real confidence. 2 Allow your internal maps to update.

The best way to do this is to exit what neuroscientists and psychologists call stimulus-response mode. Here are two ways to do so: • Engage in an activity unrelated to any ambitions or outward roles. Draw, read, think, stare at the sky, go biking... whatever it is, this is just for you.

that can help you build your best life. Let me explain. Shut Up This means don't whine; there is always someone suffering more than you. Instead, adopt a daily gratitude practice—and do it the way the science has shown to be effective (more on that below). It's especially important to avoid whining and practice gratitude when you are feeling overwhelmed or down.

I know many people are probably thinking, "So I'm supposed to suppress my feelings?" Hollis is a psychologist not a coach. He's all about acknowledging feelings. But he also is aware of what it takes to build a life that brings a person lasting joy. So stay with me. Suit Up This means be prepared.

Take the steps you can each day to arrive at your outward roles as best you are able. This sometimes looks like self-care: Get adequate sleep and don't overbook yourself. It can also look like work: Finish your assignments, be on time, keep your belongings in order, schedule your kids' activities, and so on. Homework doesn't end when we finish school. Show Up This means hold yourself accountable for your work, for how you act in relationships, and indeed for all the outward roles you take on in life.

You are in the driver's seat, so to speak. Now before you tell me all that just sounds like Boomer diligence, consider this: Our inner narratives about our self-worth, and our selfesteem, are strongly impacted from fulfilling

these outward roles.3.6 Now is that the whole picture? Of course not. There are many factors at play when it comes to how we talk and think and feel about ourselves. But this is a significant fraction of it. Alas, just telling yourself, "You are enough just as you are," won't lead you to believe it.. unless you are fulfilling these outward roles.

This is what Hollis points out. Ifyou're a member of a sports team, you show up to practice and work hard, with the subsequent feeling of pride in yourself. If you're a parent, you work hard at doing right by your child, with the subsequent satisfaction of having given that your all. Holding ourselves accountable to our chosen roles also leads us into a state of thankfulness for the opportunity to perform well in them. Hollis explained to me that this state is the most important aspect of the practice.

We choose a role, we commit to it, we do well, and then we are lifted into gratitude. Giving our roles our best each day becomes an act of gratitude—the act of living our life becomes a devotional practice of being thankful. Sound corny? Perfect! Because it's not about saying it.

It's about the actions associated with each of the sentences. Let's get to it. Shut Up, Redefined At first glance, you might think that "shut up" means stop complaining. But what Hollis is after here is replacing complaining with something more beneficial. And the suggestion is more nuanced than simply "feel grateful." Rather, try spending a short amount of time each day thinking about something that causes you to feel grateful that you would not be aware of without some effort.

This is not a simple gratitude list.

It's about searching for something we would otherwise be unaware of or take for granted. Take a minute, or five. The data tell us this dedicated effort is positively transformative for your health, 3.7 and more. In many ways, gratitude is a lot like hydration, but for your emotional well-being. Its importance for basic daily functioning is overlooked.

People think of it as an extra, but it's foundational to high performance. Of all the things I've covered on my podcast, this is the tool that surprised me the most in terms of its potency. Thinking back to what we now know about thoughts—what they are and how they grow and compete with one another—it makes sense. When we are seeding thoughts of gratitude, we suppress negative thoughts. We also run simulations from a place of security and stability, rather than a place of lack.

Much more feels possible from that place. So forget the idea that if you do a gratitude practice daily you'll become complacent. Actually, the opposite happens. Even a short gratitude practice of a few minutes has been shown to elevate motivation and mood, and can even improve memory recall and cognitive ability.3-$ Gratitude tends to increase other positive feelings and states, like happiness and optimism too.3.2 And there is evidence that this link is causative. In a 2003 study by psychologists Robert Emmons and Michael McCullough, 192 undergraduates were randomly assigned to one ofthree groups, each of which was asked to reflect on the past week by writing about (1) up to five things they were grateful for; (2) up to five things that were a hassle (a dirty kitchen, doing poorly on a test, etc.); or (3) five events that affected

them in some way in the past week (attending a concert, cleaning a closet, and so on).40 All three groups also rated their mood, any physical symptoms they were experiencing, such as headaches, congestion, sore muscles, nausea, skin irritation, poor appetite, or other minor issues, and how they reacted to any offers of help over the past week. A second study followed the same design but with daily writing instead of weekly. And a third study randomly assigned people with neuromuscular disease to either the first writing group or a control group that did only the selfrating part of the study. All three studies—weekly reflecting, daily reflecting, and reflecting among people coping with a serious illness—showed that deliberately exercising gratitude was associated with more optimism, fewer minor health issues, more exercising, a greater tendency to lend emotional support to others, and other benefits. Among the participants with neuromuscular disease, those in the gratitude writing group slept better and felt more connected to others.

Importantly, the data analysis showed that these outcomes were the specific result of inducing gratitude in oneself and not a side effect of the study as a whole. Gratitude is also good for the

heart. One study of eighty-six adults looked at changes in blood pressure and heart rate in response to a stimulus and alongside participants' responses to questions intended to gauge their level of gratitude.4J The researchers found no connection between trait gratitude and cardiovascular reactivity. However, participants with higher levels of state gratitude tended to not experience the same spikes in blood pressure that the other

feel gratitude, psychologist Robert Emmons, professor emeritus at University of California, Davis, and a pioneer in the research of gratitude and positive psychology, tells us how. He found that gratitude comes from a two-step cognitive process: (a) recognizing that one has obtained a positive outcome [in some area of life], and (b) recognizing that there is an external source for this positive outcome. Cultivating gratitude requires commitment. So Hollis's recommendation of pausing each day to acknowledge what we are grateful for is a logical starting point. But weaving together some compelling research suggests ways to make this protocol even more effective.

First, pioneering neuroscientist Antonio Damasio and colleagues investigated brain regions associated with gratitude by asking participants to listen to stories from genocide survivors who received lifesaving assistance from strangers and to imagine how they would feel about that care, rating their level of gratitude for each incident of assistance.46 The researchers found that participants documented feelings of gratitude correlated with brain activity in the anterior cingulate cortex and the medial prefrontal cortex, regions associated with values, morals, and theory of mind (which refers to the ability to understand what another person might be thinking). The approach taken in this study is important. Note that the participants here were not thinking about their own lives. They were putting themselves in other people's shoes.

The final clue here comes from a study in which participants wrote letters of gratitude to their coworkers, unbeknownst to that coworker.4Z When the coworkers received these letters, researchers were on call. They took brain images while the coworkers listened to what had been written about them. Like in the prior study, hearing these feelings of gratitude correlated with prefrontal cortex activation, suggesting that the recipients were experiencing their own gratitudes All of these findings point toward the importance of practicing gratitude. Make it nonnegotiable for a few minutes each day. Try Hollis's method.

Or put yourself in the mindset of others experiencing feelings of gratitude through story or solicit examples of when your friends, family, and colleagues felt grateful for you. You can also seek out stories that are powerful for you—such as in documentaries or in books or movies—where you observe someone sharing gratitude for actions performed by others. This approach is likely to be as effective as listing what you personally are grateful for, as is often recommended in journaling practices. It is aligned with how our brains link observing gratitude to our internal sense of reward.4-g We are clearly wired to feel better from observing and receiving genuine gratitude. Suit Up, Redefined "Suit up" may sound somewhat, well, obvious: We all know to shower, get dressed for school or work, pack our lunch and perhaps the kids' too, and get on with the day.

But this isn't about any

of that. It's about what neuroscientists call task switching. Task switching is an essential aspect of role fulfillment but is rarely discussed. Moving our attention from one chore or job or activity to another is fundamental to being effective in whatever outward roles we perform. Good task switchers are more focused and they are calmer when things get busy.42 They are

often the kind of people about which others say, "I don't know how they do it all, much less so gracefully...

Neuroscience has a lot to say about task switching, and it has everything to do with our earlier discussion about thoughts. As we spend more time doing a task, and the internal simulations around that task get more and more elaborate, a greater and greater switch cost accumulates.50 That means that it gets harder to move on from that task, and those internal simulations, to a new task, and new internal simulations. Switch costs are real and significant, reducing our mental and physical performance by anywhere from 15 to 40 percent. Sometimes it's a slowing of response times, other times it's a reduction in accuracy, or both.5-J Think of it this way: Doing something for a long time gets you deep into a neural trench, so to speak, and getting out of that trench requires more time and more energy. Some people seem to intuitively know about this aspect of brain function, and always add a few minutes between tasks to get out of a neural trench.

Others of us make the false assumption that we can hop between tasks and be at our best. You know who doesn't? Athletes, cognitive athletes (like professional chess players), and creative artists. They are

famed for their stereotyped rituals around both practice and competitions. They know how to "get into frame," "reset between points," and on and on.52 Six-time Mr. Olympia winner Dorian Yates—known for his brief but extremely intense training (which, by the way, was based in large part on Mike Mentzer's principles I described in chapter 2)—told me he ironed his clothes prior to every workout... not just for the appearance but as a mental preparation ritual for what was to come. He was actively steering his mind.53 Loads of books on high performance have delved into such practices and found that, when it comes to routines to harness our mental and physical resources so we can apply them with the utrnost focus and effectiveness, what works best is what can be ritualized, not necessarily the specific thoughts or actions.54 It's not that you take three Slow breaths emphasizing the exhale to Slow your heart rate or intentionally blink fast five times before you sit down to work, it's that you do something that you can do most anywhere, anytime and that you have decided is your ritual. Pick one, and design your own task-switching ritual.

Studies across a range of sports and cognitive tasks, like public speaking or test taking, show that stereotyped physical actions are the most effective task-switching rituals, as opposed to just mental mantras or self-talk. Think of the way a baseball player taps the bat on home plate before the first pitch. These may "shake off" the neural residue of the thoughts and actions related to your prior activity by shutting down the specific circuits involved in it. This is a distributed phenomenon, which means it does not happen in

to our cognitive, emotional, and physical capabilities.5-g And yes, many of the things that you may already do—taking a few long exhale breaths as you return home at the end of the day, or before getting into work mode when you arrive at the omce or into workout mode when you step inside the gym—are about this whole notion. So make it work for you. And remember that the ritual between tasks associated with different outward roles can be exactly the same. Same for every switch. It's yours; do whatever works.

This tool, as I said above, places the locus of control inside us. It moves us away from the need for conditions outside us to always be optimal to tackle our outward roles. It puts us in charge of how we start our tasks, and gives us confidence to continue. I'm guessing that when you first read Shut Up. Suit Up.

Show Up., you probably weren't thinking about gratitude practices, or about your basal ganglia, but what makes Hollis's approach uniquely powerful is that it balances introspection with action. It's neither coddling nor harsh. And it's realistic to the demands of life we all have to meet, as well as the internal challenges most all of us feel at some point or another. I take comfort in the fact that there are science-backed tools to help foster Shut Up. Suit Up.

Show Up. We're an Oddly fortunate species. On the one hand we are born with unmatched cognitive capacity, as well as the ability, through neuroplasticity, to continually learn. No other creature alive has that and yet, because of those very gifts, we can get stuck in our own heads on big decisions before we make them or sometimes

wake up one day to realize we took the wrong path after we did. Building your best life requires making Shut Up. Suit Up. Show Up. a daily practice, just as Hollis suggests. But we have to look inward, too, to assess if we are acting in alignment with our values and our desires.

And to know, with confidence, what those values and desires are, and that they are worthy of our time and attention. There's a Clear way to do that. Here's how that is done: Take time every day to update your internal maps. The other piece of the puzzle is to make sure you know yourselfwell. Your fears, hopes, aspirations.

This kind of self-knowledge is hard to come by. The combination of life's demands and the everpresent noise of modern life can make it hard to simply know ourselves. That's where the second part of this protocol comes in. Is this necessary? Yes.

People who excessively focus on excelling in outward roles often feel unfulfilled or they end up neglecting other important areas of life. Yet those who are excessively focused on self-knowledge and selfcare—another admittedly squishy term—often put off taking action that would allow their careers and relationships to thrive. There needs to be a balance. The solution, Hollis explains, is to incorporate a second daily practice of exiting the stimulusresponse mode that our outward roles— employee, parent, spouse, et cetera—require of us, giving us a chance to direct our attention inward instead. This protocol is like leaving the highway to take a local road: You benefit from the scenery, absence of high-speed tramc, and the

insights that inevitably come from this change of pace. This pause allows us to separate what we think other people think about us from what we actually know to be true. That separation improves our awareness of what we want and clarifies what might make our lives genuinely more meaningful and more aligned with who we are. There are two sureñre ways to go about exiting the stimulus-response mode. First, engage in an activity unrelated to any ambitions or outward roles.

Draw, read, think, stare at the sky. whatever it is, take time each day for this activity. The key is that the activity doesn't involve any other people, or their expectations or projections of you, or even your dog or cat. That means social media is out; so is posting photos from this activity. This is about you and you alone. Even just ten minutes can make a meaningful difference.

Second, simply close your eyes and ask how you're feeling, and watch for what brings you joy or makes you uncomfortable. Try one, or both. Each is a way to better know who you are as an individual, and better understand what distinguishes you from others. Swiss psychiatrist Carl Jung believed that this process is a core aspect of human existence. Hollis's writings and lectures, too, often focus on the importance ofthis process as a foundation for any other

relationship we may have.

These insights might sound wishy-washy or even self-indulgent, but they are neither. Pausing from the constant need to respond to this or that stimulus gives us the chance to hear, then question, what we have internalized from others,

roles after all. Just for brief periods of time, but daily. Take a few minutes alone without inputs from others or even your phone. Spending time alone can help quiet negative voices we've internalized so that we can better hear our own inner voice. It's a skill to learn to hear your own inner voice, and understand it.

But it is one well worth cultivating. This isn't to suggest that we already have the answers to every challenge. Sometimes we need to ask for help, and encouragement, from others. Psychotherapy, for instance, can be extremely beneficial. Data show that it can improve symptoms of psychological dificulties, that these improvements remain stable for years, and that these patients use less healthcare than the population at large after completing therapy.é2 Still, for all these reasons, alone time tends to cause peaceful feelings, even for extroverts.

Researchers from the Netherlands and the UK asked 178 people to keep a daily diary for twenty-one days in which they noted how much time they spent in solitude each day, regardless of the activity they were doing, among other reflections.63 On average, days with more hours in solitude were lonelier for the participants. However, the researchers found that choice made the difference. People were happier being alone when that was what they chose. And across the entire study group, days with more alone time were less stressful. People tended to feel more themselves and more free from external pressure on days when they had comparably more solitudes Combining alone time with activities that aren't part of the daily demands of life (work,

errands, and such) adds to the benefit, because these "extra" activities also tend to be good for our mental health. In September 2023, Nature Medicine published a remarkable study by an international group of researchers in which they looked at data from five long-term studies conducted in sixteen countries enrolling more than 93,000 people.64 This analysis found that having a hobby was tied to fewer depressive symptoms, greater happiness, and a greater sense of life satisfaction. The more time spent with these pursuits, the greater these benefits. This study focused on Older adults, so one might wonder whether the benefits of a hobby are because they give people who've left the workforce something to do. But the percentage of retirees ranged from 56 percent to 99 percent among the studies included in the analysis, which means that some of the participants Still held a job; the benefits cannot be explained by the notion that the hobby is simply replacing a career.

Carving out time to pursue our personal interests is good for our health, whether we are Still in the workforce or not. We need time to get to know ourselves. Constantly reacting to external demands leaves no space for thoughts and feelings to surface into our full awareness. It can be tempting to think that who we are is what we do, who we know, where we live—that these are the ingredients of our identity. But when we exit the stimulusresponse mode, we can tune in to questions, or even answers, that have been quietly brewing but not fully acknowledged.

Hollis speaks about these pauses as moments to "recollect," as in re-collect ourselves. I would encourage you to try it.

If you find it hard to just... sit there in silence, then try a different entry point. I often turn to drawing. There is no outside demand about what I draw or how I draw it. For you, it may be playing an instrument or gardening. "But those activities have to be done correctly, and I can't introspect while doing them," you might say.

I get it. There is a format to many such activities, and there are often metrics by which to gauge your success at them. But that doesn't matter. The key thing is that what you're doing is not about occupying an outward role that others place on you; it's about just being you. It is easy to fall into the trap of thinking we don't have time for these things.

We do. There is always time, if not every day then at least a few times each week. I have an immensely demanding schedule and have for the last thirty or so years. I try to step away from my devices and stand or sit quietly for at least five minutes per day. Time with hobbies can be admittedly more sporadic, based on my work schedule, but I aim to spend at least an hour a week doing something completely unrelated to work or the routine demands of life.

By stepping away, you'll develop a closer relationship with your own inner voice. Insights and ideas inevitably surface. Not always right away, but everyone I know who does this protocol has described a moment when something related to a thought they had during solo time days prior suddenly Shot to the surface of their awareness. The idea was a Clear "go," a call to action. Or a Clear "no go," a call to halt certain patterns in their life going forward—either small or large.

Somehow what was previously suppressed or hidden from

Exercise Protocol 1 for more about walking, and Exercise Protocol 3 for more about how to know when movement tips over into cardio work). And leave the Phone at home. The point here is to give yourself some aimless time where the only variable worth measuring is your connection to yourself.

PERSONAL GROWTH PROTOCOL Resolve Inner Pain Points Through Expressive Writing

WHAT TO DO

• Write for fifteen to twenty minutes about a challenging experience that affected your life. DO this exercise four times, once per day for four consecutive days. Keep in mind: • Do not write for longer than twenty minutes. Set a timer and stop even if you feel you could keep going. • Do not worry about the quality of your writing. No one Will read this but you. • Write about what happened, what you felt at the time, what you feel now, and any lasting repercussions, however large or small, from this experience.

Be honest about everything you felt, and everything you Still feel.

HOW IT WORKS

Not everyone has experienced trauma, but if you have, I encourage you to seek out the resources to address it. Whether it happened in childhood or adulthood, trauma and the shadow it leaves can be utterly destructive to our relationship to

Social psychologist James Pennebaker developed the research for this protocol, beginning with a study in the 1980s.6.2 He and a colleague asked forty-six participants, all university students, to write about traumatic life events once a day for four days consecutively. For comparison, another group followed the same writing schedule but focused only on trivial

occurrences. The researchers collected health records and recorded moods and physical symptoms self-reported by the participants. People writing about traumatic events experienced a decline in mood and an increase in blood pressure immediately following the exercise. But over the following Six months, they had fewer visits to the university health center.

Two years later, Pennebaker followed up with another study in which fifty students did the same exercise: Write about a traumatic zo experience or trivial topic four days in a row. This time, though, the researchers did not use self-reported data. Instead, alongside measuring blood pressure and heart rates before and after the writing exercise, the researchers also collected blood samples and tested blood for two markers of immune system function. Here, again, the exercise proved beneficial to health and mood. Writing about trauma, the researchers concluded, boosted immune function, led to less physical illness, and reduced stress.

A deluge of work on journaling about dificult experiences followed. More than 1,400 papers on expressive writing are now part of the scientific literature, with interest increasing in the forty years since Pennebaker's first studiesn

A 2004 meta-analysis of nine papers confirmed the improvements in health tied to this type of expressive writing.Z2 Among the groups for which this approach has shown a benefit are adolescents, people undergoing cancer treatment, caregivers, people suffering from post-traumatic stress disorder, people experiencing depression, along with many others. Expressive writing isn't a cure. But decades of data provide undeniable evidence that it helps with mental health—and physical health. Why? What is going on under the surface that makes this simple act work?

The mechanisms behind this are multiple, but studies suggest that the most powerful lies in how this type ofwriting affects the prefrontal cortex. Creating cohesive narratives that are honest—that is, truthful, even when the experience is painful to recount— appears to stimulate this region of the brain in a way that benefits us both psychologically and physically. Let's zero in on the "honest" part of the equation. A study from 2017 shows the hardwiring of this trait to the prefrontal cortex. Researchers from the University of Zurich, the University of Chicago, and Harvard University gave the 145 study participants a cash reward every time they won a die-rolling game.Z3 Only the person who threw the die knew their own results.

The test was rigged in a way that put the chances of winning any round at 50 percent. Here's where the experiment gets interesting: During the challenge, the researchers applied transcranial stimulation (a noninvasive application of a weak electric current to the scalp) to the prefrontal cortex of half the participants.

Specifically, this electric current reached the dorsolateral region of the prefrontal cortex, an area activated when we are choosing whether or not to be honest. The group with this stimulation reported winning 58 percent of their rolls, which meant they had lied approximately 15 percent of the time. The group without this stimulation reported winning 68 percent of the time, which meant they had lied about their rolls about 37 percent of the time. The full study included more tests to gain insight into what exactly was going on here. The researchers concluded that this brain region regulates honesty specifically when lying would benefit us.

The amygdala, by contrast, is connected to stress, fear, and anxiety. People under duress tend to have heightened activity in this part of the brain.Z4 Activation of the amygdala triggers the release of adrenaline and cortisol,Z5 which are crucial for responding to challenges and processing the memory of them. That helps when we're faced with similar challenges in the future. But the overabundance of these chemicals appears to affect the immune system, decreasing our ability to mount an immune response over time. The specific journaling exercise pioneered by Pennebaker likely works because it avoids activating the amygdala and instead connects to the prefrontal cortex.

An added benefit is that any time spent with tasks that deliberately engage the prefrontal cortex may improve brain plasticity, which has a long-term benefit on cognitive function (for more on this link, turn back to chapter 6).ZZ

recover before moving to the next part of your day. I did this as described here and I don't feel any better about the past. This exercise is not designed to resolve emotional pain entirely. It has physical and psychological health benefits, but it is not a cure. Therapy or other interventions are often needed in order to work through emotional pain so that it no longer affects us negatively.

Meditation, EMDR, hypnosis, exercise, and other techniques are all widely available. I cannot say what approach Will work for you but if you are suffering from emotional pain, then do not hesitate to seek professional help.

PERSONAL GROWTH PROTOCOL Reframe Loss to Effectively Move Through Grief

WHAT TO DO

• Understand grief as a lack of reward. Over time, your brain has learned to predict certain events, like our expectation of someone's voice or smell. It is a process that draws on our expectation of a "reward" that is no longer there. • Spend ten to thirty minutes a day to deeply feel your attachment to the person, animal, or thing you are grieving. Do not try to reduce the intensity of your reaction to this attachment. Experience it fully. • Avoid what-if thoughts.

Try not to dwell on how else a situation might have gone.

HOW IT WORKS

Losing people, animals, and things is part of life. It's healthy and normal to experience grief in the wake of those events, provided it eventually wanes. What's "eventually"? That depends. The loss of a parent or sibling is devastating.

The loss of a keepsake can be, too, but in a very different way. These are obviously not the same. But no

with motivation and craving. A study led by neuroscientist and grief researcher Mary-Frances O'Connor, at UCLA, proved just that. O'Connor and her team imaged the brains of twentythree women who were actively experiencing grief, eleven of whom were experiencing a state referred to as prolonged or complicated grief,3.3 generally defined as intense, debilitating, and unrelieved grief lasting more than a year. The women viewed photographs of the deceased and words related to their death as brain images were taken using functional magnetic resonance imaging. Brain images of the women with complicated grief showed activity in pathways connected to motivation and reward during moments when the women reported

feelings of yearning.

These neural circuitries lead to the release of chemicals that make us feel good, like dopamine, serotonin, and oxytocin, keeping us motivated to obtain things that are good for us, like food and social connections. Specifically, the images showed activation of the nucleus accumbens, which releases dopamine during social interactions. This (at the time) surprising observation led the researchers to hypothesize that grief is akin to searching for a reward that never arrives. We stay motivated to keep finding it, but we don't —and can't. That makes sense to anyone who has experienced grief but until O'Connor's work, neuroscientists were unaware that grief was tied to reward anticipation circuits in the brain.

We all experience loss. This is a lesser example than the one described above, but simply for example: Anyone who's Iost a large bill knows what it's like to look in their wallet and not find

it. You can tell yourself it's just money, and you Will get over it, but the loss of it brings the sudden understanding that whatever it made possible is now gone. Translate that to more emotional losses, such as a loved one or a friend or a pet. Suddenly our neural circuitry around that attachment is no longer actionable. The pathways that used to fire are Still there, but they aren't useful anymore.

What is possible has changed. The circuit of anticipation and action resolution remains open. And that's painful. But motivation circuitries aren't the only ones at play in this state of longing. Our brains contain cells dedicated to mapping the world we move through—think of it like a sort of mental map of where things are.85 Likewise, we also map our relationships, keeping track of the ties that connect us, albeit with different brain cells.

Studies have found some overlap among neurons we use to orient ourselves in space and time and those we use to understand our relationships to other people.86 Researchers at Dartmouth College found that physical distance, proximity in time, and social closeness activate the same region of the brain, namely the inferior parietal lobeg-Z The circuitries that store information about the distance from your bed to the bathroom, from your house to the supermarket, and from your town to Paris overlap with circuitries that store information about how close another person is to you socially. Rewiring the neural circuitries that orient us with regard to the people in our lives takes time. We experience this physically often: Ifyou've ever rearranged the furniture in your living room, for example, you know that it can take time to

remember that the couch isn't where it used to be. Grieving includes a similar reorientation but, of course, with heightened emotionality. We have to get used to our loved one not being where we always found them—arriving home from work at 6:00 pm., at the other end of the phone, across from us at the dinner table. We no longer experience the rush of chemicals we once did when that happened, what our brain registered as a reward. Grief exists where these circuitries contradict with reality, what O'Connor refers to as the "Gone-But-Also-Everlasting" theory of grief.8-$ It takes time for our brains to update.

Thus grief is not only a disruption to our circuitries connected to motivation and reward, but also a dificult learning process. Avoidance makes this learning process even harder because it prevents us from incorporating new information into the map. If we are refusing to face reality, then our brains can't integrate it. As O'Connor has written, "Learning requires experience, and avoidance of things that remind the bereaved of the loved one's death or absence Will prevent accommodation of the information. Instead, allowing ourselves to really feel our attachment to a person we've Iost, a pet who's died, or even a missing object that held

sentimental value, without denying what happened, gives our brains a chance to decouple the object of our love from our current environment and our brain's expectation of being able to interact with the person (or pet) that was Iost.

This decoupling—accepting a death while also savoring the memories and the essence (meaning our internal representation of

In one study, researchers examined links between counterfactual thoughts and symptoms stemming from prolonged grief and depression among fifty-nine people suffering a recent loss. They found a strong correlation. People who engaged in what-if thinking experienced a longer period of grieving and symptoms of depression. Besides, grief is hard enough without the guilt. Psychology, philosophy, religion—all these disciplines have considered grief, tried to help us understand Why we experience it and how to live with it.

The protocol above is one of neuroscience's offerings. Understand that grief is like a craving for a reward that is no longer coming. Make space to feel the deep attachment, whether through small rituals or just finding moments in your day to pause and fully experience your feelings. This time Will allow your brain circuitries to rewire, creating memories along with new mental maps. Finally, resist engaging in counterfactual thinking, which can prolong suffering, trigger feelings of guilt, and exacerbate any depressive symptoms arising during this period.

COMMON QUESTIONS What is a healthy amount of time for grief? Psychologists tend to consider prolonged grief to be grief that is debilitating and lasts more than a year. With prolonged grief, people feel depressed and hopeless, often stop engaging socially, and may have little appetite. Not all grief is like this, but it's hard to put a time frame on when any grief should end, since the process is so personal. It's more important

to understand what healthy grieving is, rather than putting a time frame on the process. Allow yourself the time and space to be with your deep feelings of attachment and closeness and let the process unfold What I can say, though, is that one day, you might be walking along and remember someone you Iost, and you might experience that person —not just the memory of them, but them—as a completely positive presence that somehow (and I don't have any science to explain this) is fully integrated into who and where you are now, right at that moment. Meaning: You know they are gone, and yet they are Still very present. The remapping has occurred. That's happened in my life many times over.

To me, that is the end stage of the grief process, one that fortunately never subsides. It just takes some time to get there.

Before You Turn the Last Page Although this is the end of the book, I hope you see it as a beginning—the start of a new direction for your health that is rooted in a deeper understanding of the incredible thing that is the human brain and body, and how each and every one of us can improve our mental and physical health and learn new things that rewire our nervous system. As I mentioned in the introduction, each of our lives is its own kind of laboratory. There is no question that there are hardwired aspects of the human brain and body, common to us all, and thank goodness for those aspects. They are much of what the protocols in this book are based on, and Why they work. But we should all also acknowledge our individuality.

So whether you are working on improving your sleep or stress control, changing your exercise regimen or nutrition to get leaner, stronger, more mobile, or all the above, it's important that

you honor the universal aspects of human biology but also feel free to make adjustments to suit your needs. The right way to do that—the scientific approach that is—is to become a systematic observer of you. Record your results. Then raise hypotheses, ask questions, and

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